

Payton Suh | Los Angeles, CA | paytonsuh@ucla.edu | paytonsuh.com

Interested in pure mathematics, probability, and their intersections with the physical sciences and computer science.

EDUCATION

UNIVERSITY OF CALIFORNIA, LOS ANGELES — Mathematics B.S.

Expected Jun 2028

Departmental Scholars Program (concurrent B.S./M.A.): applying

Current Coursework: Algebra (110AH); Linear Algebra (115AH); Analysis (131AH), all Honors

Planned Coursework: Abstract Algebra, Analysis, Complex Analysis, Geometric Analysis, Topology, Probability Theory, Stochastic Calculus

SANTA MONICA COLLEGE — Mathematics AS-T & General Science AS

Aug 2024 – Jun 2026

Cumulative GPA: 3.85/4.00 | Dean's List 2024–2026

Relevant Coursework: Calculus I–III; Linear Algebra; Differential Equations; Engineering Physics I–III; Modern Physics; Chemistry; Programming (Python, C++, Data Structures, MATLAB); Data Science; Circuits Analysis.

EXPERIENCE

STUDENT INSTRUCTOR — SANTA MONICA COLLEGE

Jan 2026 – Present

Student Instructor

- Selected by Santa Monica College's STEM Department to provide academic instruction in Physics and Chemistry to college students at the Science Tutoring Center.
- Deliver personalized academic support through one-on-one and group sessions (10+ weekly @ 8 hours a week) to strengthen conceptual understanding and problem-solving skills in undergraduate STEM subjects.

DATA ANALYST — CENTER FOR PAIN CONTROL

Jun 2025 – Jul 2025

Accelerated Summer Internship

- Managed and maintained large-scale data systems (5,000+ entries) while coordinating with legal firms to streamline billing, legal documentation, and financial workflows.
- Collaborated cross-functionally with legal and finance teams to process subpoenas, invoices, and payment reconciliations with 99.9% (near perfect) data accuracy and compliance.

ACTIVITIES

ZrC-ARC — Zirconium Carbide Architected Refractory Coatings for NTP

Jan 2026 – Present

Co-Investigator

Oct 2026 – Present

- Lead the research and simulation side of the project, modeling grain morphology, thermal-expansion mismatch, and carbon vacancies in the ZrC lattice with LAMMPS molecular dynamics.

Principal Investigator (PI)

Jan 2026 – Oct 2026

- PI for ZrC-ARC, a student-led research project investigating the role of granular morphologies in performance-based refractory coatings for Nuclear Thermal Propulsion.
- Working alongside Prof. Daniel R. Mumm at the University of California, Irvine's Materials Research Institute. Winner of national proposal writing competition and selected for \$10,000 USD in seed funding from the NASA L'SPACE Program.

VEX ROBOTICS

Jun 2023 – Present

Mentor and Lead Builder (Teams 11444X, 334R)

- Mentored team to VEX Worlds Division Semifinals and California Region 4 State's Tournament; guided mechanical design, systems integration, and autonomous programming via robotic design optimization, sensor calibration, and control algorithm development to improve competition game win-rate performance by ~25% (11444X).
- Awards before mentorship: Achieved Haunted Signature Event Championship (11444X) and California Region 4 Create Award (334R) for innovation, strategy, and engineering excellence.

BACK TO SCHOOL TOGETHER

Jul 2025 – Present

Vice President of California

- Raised over \$1,200 in funding to support STEM education initiatives for underserved students in Mexico and Guatemala through strategic fundraising and community engagement, coordinating volunteer outreach and managing international outreach programs to expand STEM accessibility, educational equity, and sustainable impact.

PROJECTS

LIGHT CURVE ANALYSIS

Oct 2025 – Jun 2026

- Wrote Python code and utilized machine learning algorithms to extract, decompose, and analyze stellar light curves from public databases to identify light-emission abnormalities from distant stars.
- Co-authored a research submission documenting the discovery of a previously unreported variable star exhibiting frequent, high-energy flares.

EXTRACURRICULARS

HOLA SYMPHONY ORCHESTRA & TORRES CHAMBER ENSEMBLE

Aug 2018 – Jun 2024

Violin I

- Played 1st violin for the Youth Orchestra of LA's HOLA Symphony Orchestra and Torres Chamber Ensemble. Through weekly rehearsals, sectionals, and performances: refined my violin's voice, used effective peer communication, and adapted to the fast-paced environment of a high-level orchestra.

ELITE TAEKWONDO CENTER — KOREATOWN

Aug 2017 – Jun 2024

III Black Belt + Instructor

- Kukkiwon certified 3rd degree black belt in Taekwondo and have trained for over a decade.
- For two years, instructed classes to a wide range of belt colors, teaching fundamentals, poomsae form, and sparring techniques to youth students. The dojo taught respect, communication, and physical discipline through mentorship.

ADDITIONAL

Technical Skills: MATLAB, Simulink, Python, C++, LaTeX, SQL, SolidWorks, CAD Modeling, LAMMPS, Data Analysis, Microsoft Excel, Data Visualization, Circuit Analysis, Statistics, Engineering Design, Project Management, Technical Documentation

Languages: English (Native), Korean (Native), Japanese (Advanced)

Interests: Boulderling (V7, 3+ times weekly), Billiards (8-ball and 9-ball, 4+ times weekly), Cooking, Photography